

Notations :

- 1.Options shown in green color and with ✓ icon are correct.
- 2.Options shown in red color and with ✗ icon are incorrect.

Change Font Color :	No
Change Background Color :	No
Change Theme :	No
Help Button :	No
Show Reports :	No
Show Progress Bar :	No

AI

Section Id :	640653139236
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	11
Number of Questions to be attempted :	11
Section Marks :	32
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653390094
Question Shuffling Allowed :	No

Question Number : 1 Question Id : 6406532136623 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : AI: SEARCH METHODS FOR

PROBLEM SOLVING (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406536958979. ✓ Yes

6406536958980. ✗ No

Sub-Section Number :

2

Sub-Section Id :

640653390095

Question Shuffling Allowed :

No

Question Number : 2 Question Id : 6406532136624 Question Type : MSQ

Correct Marks : 0 Max. Selectable Options : 0

Question Label : Multiple Select Question

**ASK FOR PRINTED
GRAPH SHEETS
6 PAGES TWO-SIDE PRINT**

Options :

6406536958981. ✓ Printed graph sheets were provided on time.

6406536958982. ✗ Printed graph sheets were provided late.

6406536958983. ✗ Printed graph sheets were not provided.

6406536958984. ✓ I used the graph sheets.

6406536958985. ✗ I did not use graph sheets.

Sub-Section Number :

3

Sub-Section Id :

640653390096

Question Shuffling Allowed :

No

Question Id : 6406532136625 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (3 to 4)

Question Label : Comprehension

SEARCH

Consider a water-jug puzzle with three jugs a, b and c of capacities 5L, 3L and 2L, respectively.

A state, encoded as a digit string ABC, denotes volume of water in jugs a, b and c, respectively.

For example, state 212 denotes that jugs a, b and c contain 2L, 1L and 2L of water, respectively.

The allowable moves (actions) are

xETy: pick up jug x and empty it into jug y and jug y is also topped up,
xEy: pick up jug x and empty it into jug y and jug y is still not topped up,
xTy: pick up jug x and top up jug y and jug x is still not emptied out.

For example: 212 $\xrightarrow{\text{cEa}}$ 410 $\xrightarrow{\text{aTb}}$ 230 $\xrightarrow{\text{bETa}}$ 500.

MoveGen takes a state ABC and returns a set of neighbours that are one move away from ABC.

For example, $\text{MoveGen}(212) = \{ 032, 302, 410, 230 \}$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 3 Question Id : 6406532136626 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

Let 320 be the start state and 401 be the goal state, find the shortest path from start to goal. Enter the path starting with 320 as a comma separated list of states.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 320,302,500

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

320,122,131,401

Question Number : 4 Question Id : 6406532136627 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

Let 320 be the start state and 401 be the goal state, find the sequence of moves that produce the shortest path from start to goal. Enter the sequence of moves as a comma separated list.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: aEb,aTc,bETa,cETb

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

aTc,cTb,bEa

Sub-Section Number : 4

Sub-Section Id : 640653390097

Question Shuffling Allowed : No

Question Id : 6406532136628 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (5 to 7)

Question Label : Comprehension

SEARCH ALGORITHMS

Answer the given subquestions.

Sub questions

Question Number : 5 Question Id : 6406532136629 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Which of the following variants of Depth First Iterative Deepening (DFID) is guaranteed to find the shortest path if one exists?

Options :

6406536958988. ✖ DFID that inspects only new nodes.

6406536958989. ✖ DFID that inspects new as well as open nodes.

6406536958990. ✔ DFID that inspects new as well as closed nodes.

6406536958991. ✖ None of these.

Question Number : 6 Question Id : 6406532136630 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Given a finite state space with edge costs that may or may not be Euclidean and a heuristic function whose properties are not known, which of the following algorithms are suitable for finding the optimal path?

Options :

6406536958992. ✖ Depth First Search

- 6406536958993. ✖ Breadth First Search
- 6406536958994. ✔ Dijkstra's algorithm
- 6406536958995. ✔ Branch and Bound Search
- 6406536958996. ✖ A*
- 6406536958997. ✖ WA* for some w
- 6406536958998. ✖ Sparse Memory Graph Search (SMGS)

Question Number : 7 Question Id : 6406532136631 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

If w is set to zero then wA* algorithm will _____ .

Options :

- 6406536958999. ✖ behave like Best-First Search
- 6406536959000. ✔ behave like Dijkstra's algorithm
- 6406536959001. ✔ always find the optimal path
- 6406536959002. ✖ sometimes find a suboptimal path

Sub-Section Number : 5

Sub-Section Id : 640653390098

Question Shuffling Allowed : No

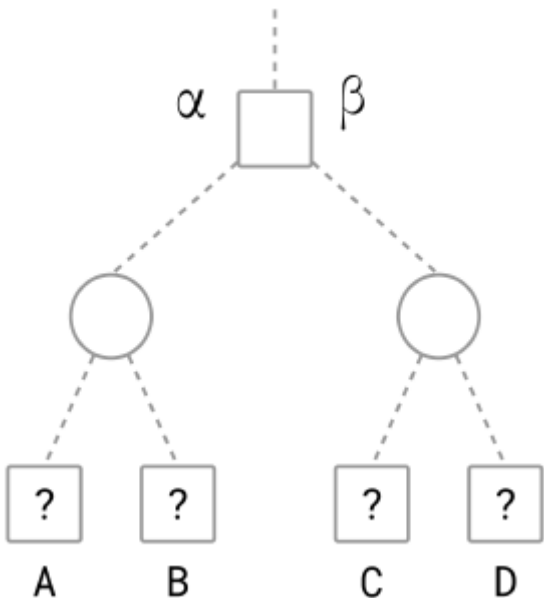
Question Id : 6406532136632 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (8 to 11)

Question Label : Comprehension

GAMES: ALPHA-BETA

Consider a game tree with the root node as MAX, where an arbitrary path from the root reaches the subtree shown in the figure.



Each leaf node A to D takes a **UNIQUE EVAL VALUE** from the set {1, 2, 3, 4}.

Alpha-Beta algorithm is entering the subtree with $\alpha=2$ and $\beta=4$;
find an optimal eval assignment (for nodes A to D) that maximizes the number of leaves pruned;
find the minimax value, the type of cuts and the leaves pruned for that assignment.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 8 Question Id : 6406532136633 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

Enter the optimal eval assignment (evals of nodes A to D) in the text box.

Enter a comma separated list of evals.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 1,2,3,4

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

1,3,2,4

1,4,2,3

2,3,1,4

2,4,1,3

Question Number : 9 Question Id : 6406532136634 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

The minimax value of the subtree for the optimal eval assignment is _____ .

Enter an integer.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2

Question Number : 10 Question Id : 6406532136635 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

Enter the number of alpha-cuts followed by the number of beta-cuts as a comma separated list.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 1,2

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

2,0

Question Number : 11 Question Id : 6406532136636 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

Enter the label of leaf nodes pruned by Alpha-Beta algorithm, or enter NIL if no leaves were pruned.

Enter a comma separated list of labels (A to D), or enter NIL.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: W,X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

B,D

D,B

Sub-Section Number :

6

Sub-Section Id :

640653390099

Question Shuffling Allowed :

No

Question Id : 6406532136637 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (12 to 13)

Question Label : Comprehension

GAMES: SSS STAR

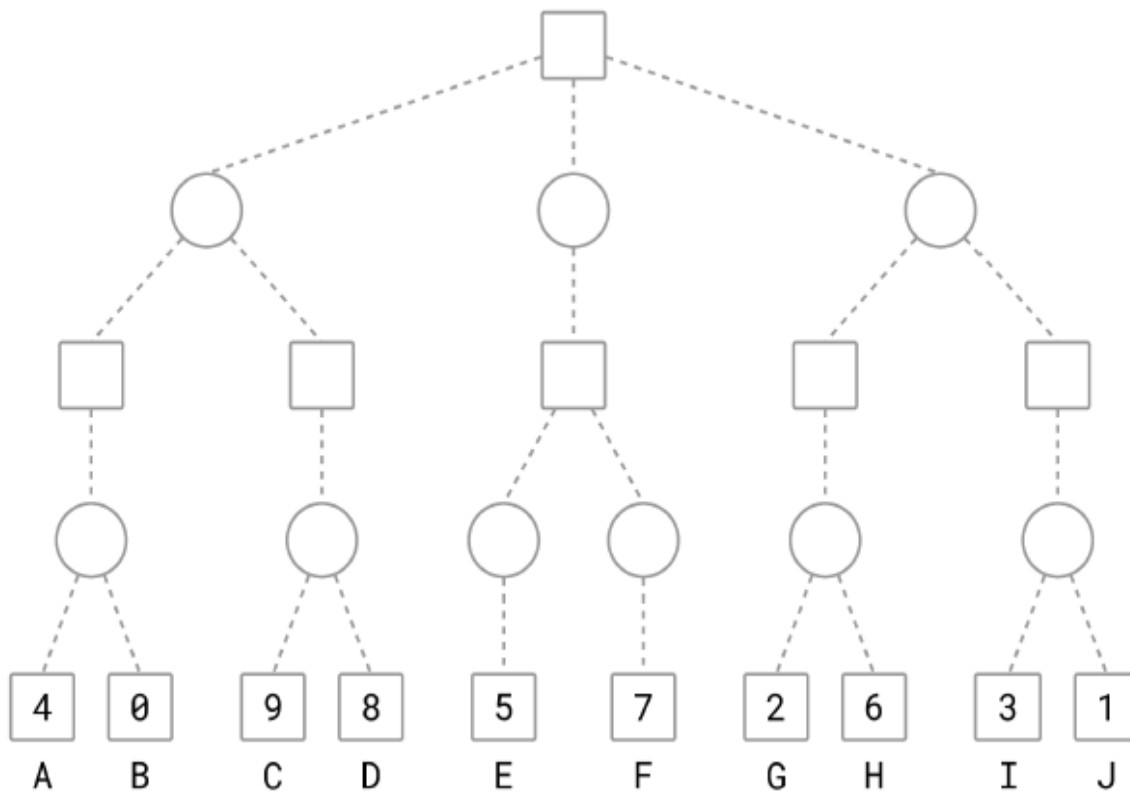
The figure shows a game tree with evaluation function values at the horizon nodes.

The horizon nodes are labeled from A to J.

Where applicable, use these labels in short answers.

Tie-breaker:

When several nodes carry the same best cost then select the deepest node, if tie persists then select the leftmost of the deepest nodes to break the tie.



Run SSS* algorithm on the game tree, then answer the sub-questions.

Sub questions

Question Number : 12 Question Id : 6406532136638 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

Find the horizon nodes that are assigned SOLVED status by SSS* algorithm. Enter the labels of those nodes in the textbox, or enter NIL if no nodes are assigned SOLVED status.

Enter node labels as a comma separated list in **alphabetical** order.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Answers Case Sensitive : No
Text Areas : PlainText
Possible Answers :

A,E,F,G

Question Number : 13 Question Id : 6406532136639 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

Find the SOLVED horizon nodes that are pruned from the queue by SSS* algorithm, i.e., SOLVED horizon nodes removed from the queue when a MAX-ancestor is SOLVED. Enter the labels of those nodes in the textbox, or enter NIL if SOLVED nodes were never pruned.

Enter node labels as a comma separated list in **alphabetical** order.
NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Answers Case Sensitive : No
Text Areas : PlainText
Possible Answers :

A,E,G

Sub-Section Number : 7
Sub-Section Id : 640653390100
Question Shuffling Allowed : No

Question Id : 6406532136640 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (14 to 16)

Question Label : Comprehension

PROBLEM DECOMPOSITION

The figure shows an AND-OR decomposition of problem S into smaller problems. The nodes are uniquely identified by labels (S, A, B, C, ...).

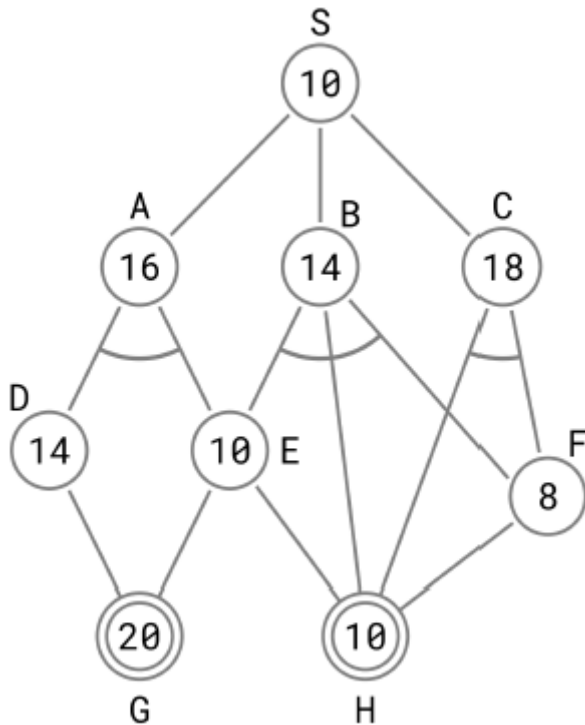
Each node shows the heuristic estimate of the cost of solving that node.
Nodes shown in double lines are primitive nodes and their values are actual costs.

A primitive node is added to the graph, with SOLVED status, when its parent is expanded. And therefore, a primitive node is never expanded.

The cost of each edge is 2 units.

Tie-breaker 1: If several nodes have the same cost then break the tie using node labels.

Tie-breaker 2: For AND nodes, select the unsolved branch with the highest cost.



Use AO* algorithm to solve S, then answer the sub-questions.

Sub questions

Question Number : 14 Question Id : 6406532136641 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

List the nodes expanded by AO* algorithm. List the nodes in the order they are expanded. Observe that primitive nodes are not expanded.

Enter a comma separated list of node labels.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: S,X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

S,B,A,C,F

B,A,C,F

Question Number : 15 Question Id : 6406532136642 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

For each node expanded by AO* algorithm, determine the value propagated to the start node S. Enter the values of S as a list.

Enter a comma separated list of numbers.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 12,42,17

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

16,18,20,24,28

18,20,24,28

Question Number : 16 Question Id : 6406532136643 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

What can you conclude about the given AND-OR decomposition?

Options :

6406536959011. ✓ The heuristic is admissible.

6406536959012. ✗ The heuristic is inadmissible.

6406536959013. ✗ The heuristic is sometimes admissible and sometimes inadmissible.

Sub-Section Number : 8

Sub-Section Id : 640653390101

Question Shuffling Allowed : No

Question Id : 6406532136644 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

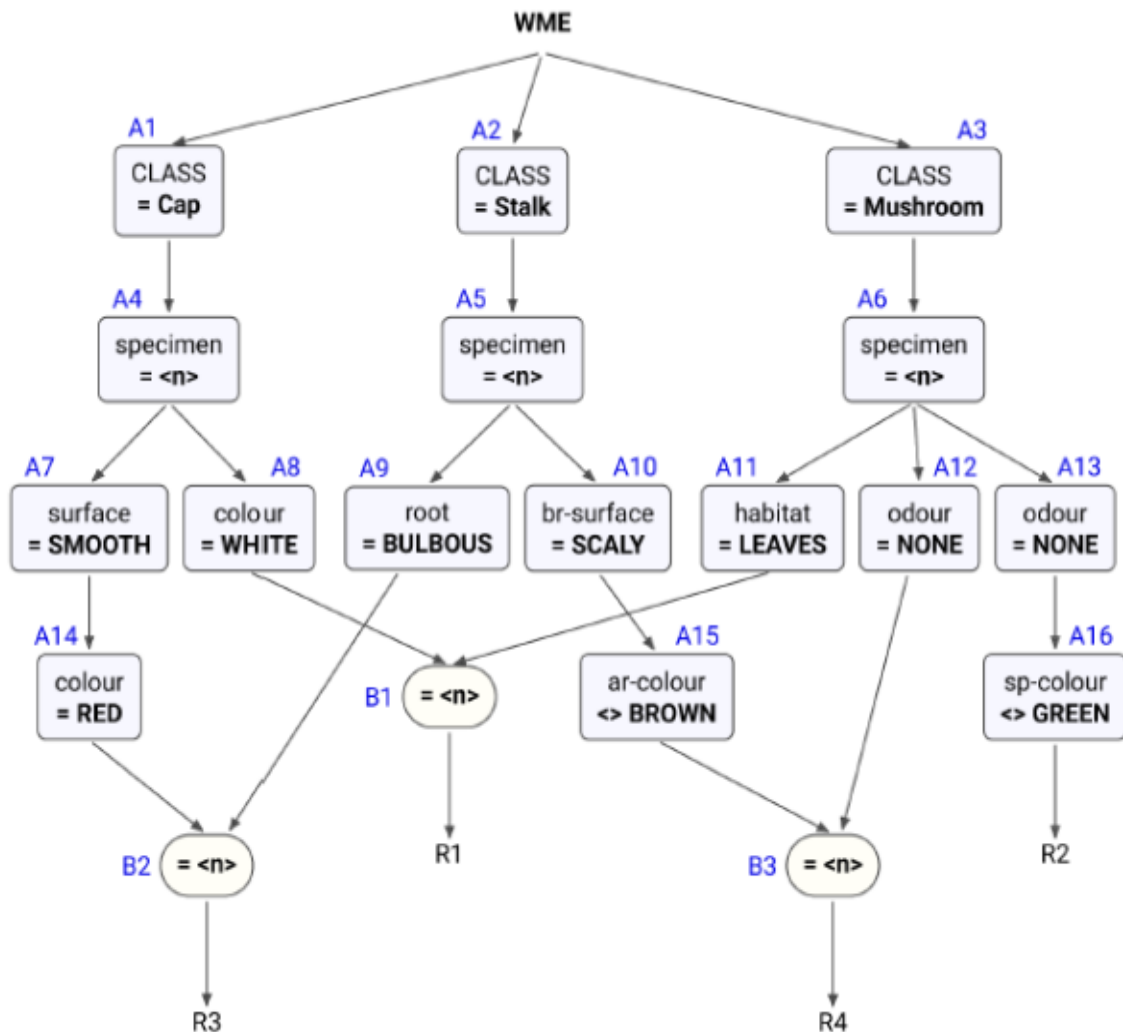
Question Numbers : (17 to 19)

Question Label : Comprehension

RULE BASED EXPERT SYSTEMS

A part of the Rete Net that classifies mushrooms (as edible or poisonous) is shown in the figure. The labels A1, A2, ..., A10, A16, ..., B1, B2, B3, R1, ..., R4 uniquely identify the nodes in the network. When required, use the above label ordering to **break ties** and to enter short answers.

Note: "attribute <> x" is true when that attribute exists and its value is not equal to x.



Run the Rete algorithm for the Working Memory shown below, the WMEs are in timestamp order. Assume that WMEs reside at appropriate Alpha nodes, and the Beta nodes point to WMEs residing in Alpha nodes.

201. (Mushroom ^specimen B43 ^odour ALMOND ^sp-colour BROWN)
202. (Stalk ^specimen B43 ^br-surface SMOOTH)
203. (Mushroom ^specimen K13 ^odour NONE ^habitat LEAVES)
204. (Cap ^specimen A23 ^colour RED ^surface SMOOTH)
205. (Mushroom ^specimen B37 ^odour NONE ^habitat LEAVES)
206. (Cap ^specimen B37 ^colour WHITE ^surface SMOOTH)
207. (Stalk ^specimen A23 ^root BULBOUS ^ar-colour WHITE)
208. (Stalk ^specimen K13 ^br-surface SCALY ^ar-colour WHITE)
209. (Mushroom ^specimen A23 ^odour NONE ^sp-colour WHITE)

For each WME identify its location (node label) in the Rete Net, and prepare the conflict set for the first cycle, then answer the given subquestions.

Sub questions

Question Number : 17 Question Id : 6406532136645 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following rule-data tuples are in the conflict-set?

Options :

6406536959014. ✓ R1,205,206

6406536959015. ✓ R2,209

6406536959016. ✓ R3,204,207

6406536959017. ✓ R4,203,208

6406536959018. ✗ R1,203,206

6406536959019. ✗ R4,205,208

Question Number : 18 Question Id : 6406532136646 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

If the Inference Engine uses **Specificity** as the conflict resolution strategy then which of the following rule-data tuples will qualify?

Options :

6406536959020. ✗ R1,205,206

6406536959021. ✗ R2,209

6406536959022. ✓ R3,204,207

6406536959023. ✓ R4,203,208

6406536959024. ✗ R1,203,206

6406536959025. ✗ R4,205,208

Question Number : 19 Question Id : 6406532136647 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

If the Inference Engine uses **Recency** as the conflict resolution strategy then which of the following rule-data tuples will qualify?

Options :

6406536959026. ✗ R1,205,206

6406536959027. ✓ R2,209

6406536959028. ✗ R3,204,207

6406536959029. ✗ R4,203,208

6406536959030. ✗ R1,203,206

6406536959031. ✗ R4,205,208

Sub-Section Number :

9

Sub-Section Id :

640653390102

Question Shuffling Allowed :

No

Question Id : 6406532136648 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (20 to 22)

Question Label : Comprehension

AUTOMATED PLANNING 1

Answer the given subquestions.

Sub questions

Question Number : 20 Question Id : 6406532136649 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider actions a and b and two **feasible** orderings (a then b) and (b then a). Which of the following conditions (each taken independently) will produce different outcomes for each ordering?

Options :

6406536959032. ✓ P is in pre(a) and Q is in del-effects(b).

6406536959033. ✓ P is in add-effects(a) and also in del-effects(b).

6406536959034. ✗ None of these.

Question Number : 21 Question Id : 6406532136650 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

In planning graphs constructed by GraphPlan, actions a and b in layer n are mutex _____ .

Options :

6406536959035. ✗ only if every P in pre(a) is mutex with every Q in pre(b)

6406536959036. ✗ only if every P in pre(a) is present in del-effects(b)

6406536959037. ✓ if P is in pre(a) and also in del-effects(b)

6406536959038. ✓ if P is in add-effects(a) and also in del-effects(b)

Question Number : 22 Question Id : 6406532136651 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

In planning graphs constructed by GraphPlan, which of the following are true?

Options :

6406536959039. ✓ If actions A and B are non mutex in a layer n then it will remain non mutex in all future layers.

6406536959040. ✗ If actions A and B are non mutex in a layer n then it can become mutex in

some future layer.

6406536959041. ✖ If actions A and B are non mutex in a layer n then it must be mutex in a past layer.

6406536959042. ✖ None of these.

Sub-Section Number :	10
Sub-Section Id :	640653390103
Question Shuffling Allowed :	No

Question Id : 6406532136652 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (23 to 29)

Question Label : Comprehension

AUTOMATED PLANNING 2

The domain description of a Blocks World with a single one-armed robot is given below.

PREDICATES

armEmpty	The arm is not holding any block, it is empty.
holding(X)	The arm is holding X.
onTable(X)	X is on the table.
clear(X)	X has nothing above it, it is clear.
on(X,Y)	X is directly placed on Y.

OPERATORS

Pickup(X): pick up X from the table.

Preconditions: { armEmpty, clear(X), onTable(X) }

Add Effects : { holding(X) }

Del Effects : { armEmpty, onTable(X) }

Putdown(X): place X on the table.

Preconditions: { holding(X) }

Add Effects : { armEmpty, onTable(X) }

Del Effects : { holding(X) }

Unstack(X,Y): pick up X that is directly sitting on Y.

Preconditions: { armEmpty, clear(X), on(X,Y) }

Add Effects : { clear(Y), holding(X) }

Del Effects : { armempty, on(X,Y) }

Stack(X,Y): place X directly on top of Y.

Preconditions: { holding(X), clear(Y) }

Add Effects : { armEmpty, on(X,Y) }

Del Effects : { holding(X), clear(Y) }

NOTE: variables X and Y must bind to different objects.

Consider the following planning problem.

Start State: { onTable(A), onTable(B),
 on(D,A), on(E,B),
 clear(D), clear(E),
 holding(C)
 }

Goal Description: { on(C,D), on(B,C) }

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 23 Question Id : 6406532136653 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are **applicable** actions for FSSP?

Options :

6406536959043. ✓ Putdown(C)

6406536959044. ✗ Stack(B,C)

6406536959045. ✓ Stack(C,D)

6406536959046. ✓ Stack(C,E)

6406536959047. ✗ Unstack(D,A)

6406536959048. ✗ Unstack(E,B)

Question Number : 24 Question Id : 6406532136654 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are **relevant** actions for BSSP?

Options :

6406536959049. ✗ Putdown(C)

6406536959050. ✓ Stack(B,C)

6406536959051. ✓ Stack(C,D)

6406536959052. ✗ Stack(C,E)

6406536959053. ✗ Unstack(D,A)

6406536959054. ✗ Unstack(E,B)

Question Number : 25 Question Id : 6406532136655 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

In the planning graph, which of the following are mutex action pairs in layer 1?

Options :

6406536959055. ✗ Stack(C,D) and nop for armEmpty

6406536959056. ✗ Stack(C,D) and nop for clear(E)

6406536959057. ✓ Stack(C,D) and nop for holding(C)

6406536959058. ✓ Stack(C,D) and Putdown(C)

Question Number : 26 Question Id : 6406532136656 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

In the planning graph, which of the following are mutex proposition pairs in layer 1?

Options :

6406536959059. ✖ armEmpty and clear(E)

6406536959060. ✔ armEmpty and holding(C)

6406536959061. ✖ on(C,D) and clear(E)

6406536959062. ✔ on(C,D) and onTable(C)

Question Number : 27 Question Id : 6406532136657 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

In the planning graph, which of the following are **applicable** actions in layer 2?

Options :

6406536959063. ✖ Pickup(C)

6406536959064. ✔ Putdown(C)

6406536959065. ✔ Unstack(D,A)

6406536959066. ✔ Unstack(E,B)

Question Number : 28 Question Id : 6406532136658 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

For the given planning problem, does a plan exist?

Options :

6406536959067. ✖ Yes

6406536959068. ✔ No

6406536959069. ✖ Cannot be determined

Question Number : 29 Question Id : 6406532136659 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

For the given planning problem, the length of the plan found by GraphPlan is _____ .

Enter an integer or enter NIL if GraphPlan cannot find a plan.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 12

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

NIL

Sub-Section Number :

11

Sub-Section Id :

640653390104

Question Shuffling Allowed :

No

Question Id : 6406532136660 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (30 to 34)

Question Label : Comprehension

CONSTRAINT SATISFACTION

Consider a CSP over 3 variables A, B, C where the domains and constraints are:

$$D_A = D_B = D_C = \{1, 2, 3, 4\}$$

$$R_{AC}: A = \text{mod}(2C, 3)$$

$$R_{BA}: B = \text{mod}(3A, 4)$$

$$R_{CB}: C = \text{mod}(2B, 3)$$

where, $\text{mod}(n,d)$ returns the remainder after dividing n by d , for example, $\text{mod}(8,3)=2$, $\text{mod}(12,4)=0$, $\text{mod}(9,4)=1$.

Compute the three relations to match the domain constraints, then draw the constraint graph and matching-diagram, and then answer the sub-questions.

Sub questions

Question Number : 30 Question Id : 6406532136661 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following tuples occur in R_{AC} ?

Options :

6406536959071. ✓ (1,2)

6406536959072. ✓ (2,1)

6406536959073. ✓ (2,4)

6406536959074. ✗ (4,2)

Question Number : 31 Question Id : 6406532136662 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following tuples occur in R_{BA} ?

Options :

6406536959075. ✓ (1,3)

6406536959076. ✓ (2,2)

6406536959077. ✓ (3,1)

6406536959078. ✗ (4,2)

Question Number : 32 Question Id : 6406532136663 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Is the given CSP arc-consistent?

Options :

6406536959079. ✗ Yes

6406536959080. ✓ No

6406536959081. ✗ Cannot be determined

Question Number : 33 Question Id : 6406532136664 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

If the given CSP is not already arc-consistent, then make it arc-consistent, and then check if the resulting network is path consistent.

Options :

6406536959082. ✓ It is path consistent.

6406536959083. ✗ It is not path consistent.

6406536959084. ✗ Path consistency does not apply because the constraint graph has a cycle.

Question Number : 34 Question Id : 6406532136665 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

Does the given CSP have a solution?

Enter the solution for the variables A,B,C as a comma separated list.

Enter NIL if there is no solution.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 1,2,3

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

